RAG-Assisted Small Language Models for Domain-Level Reasoning

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Abstract - **The increasing demand for generative AI solutions has made privacy, efficiency, and reliability crucial requirements, especially in sensitive industries where incorrect or unverified outputs can lead to major risks. While large-scale models such as GPT-4 deliver strong performance, their computational cost and dependence on cloud infrastructure often make them unsuitable for highly regulated or offline deployment scenarios. Small Language Models (SLMs) offer a practical alternative due to their lower hardware requirements and ability to be fully self-hosted, but the trade-off is reduced accuracy and weaker reasoning capabilities. In this work, we explore whether Retrieval-Augmented Generation (RAG) can meaningfully improve SLM performance and make these models more viable for real-world industrial adoption. We integrate RAG into two open-source models, LLaMa 3.1 8B and LLaMa 3.2 3B running locally via the Ollama inference engine. Their performance is evaluated using the MMLU Pro benchmark, targeting domains where hallucination and factual grounding present real challenges. To support retrieval, we constructed domain-specific knowledge bases through synthetic data generation using the grounded Gemini API and targeted legal web scraping, covering areas such as biology, virology, computer security, and common law. Retrieved passages are stored in a vector database and supplied as supporting context during inference, allowing the models to base their reasoning on verifiable information. The results show clear improvements in accuracy across several domains, demonstrating that RAG can help reduce hallucinations and enhance reliability without retraining. These findings suggest that with carefully curated data and minimal resource overhead, small models can attain performance closer to that of much larger systems while maintaining full data control and regulatory compliance. Overall, this research highlights a feasible and scalable path toward deploying domain-adapted AI locally in environments where privacy and resource constraints are critical.**

Keywords: LLM, RAG, MMLU Pro, LLaMa, Generative AI

I. Introduction

As the trend towards LLMs is increasing day by day, they are becoming a major platform in almost every sector of the world. Be it health, education or economy related fields, LLMs have proved to be a top trending tool used across these fields. They have evolved in major complex tasks such as question answering, text summarization and multilingual support, thus revolutionizing natural language processing to a great extent. LLMs models such as GPT-4, Claude and Gemini are pushing the boundaries of generative AI [1]. Besides such usefulness in the NLP field, LLMs are often hindered by the limitations of cost, energy consumption and privacy. In order to overcome these limitations of LLMs, Small Language Models (SLMs) are introduced. These models have less than 10 billion parameters that can run on local hardware. Although SLMs excel with LLMs in cost, energy consumption and privacy but due to their limited training capacities, they produce less accurate results in computation tasks.

The aim of this paper is to analyze whether Retrieval-Augmented Generation (RAG) [2] can significantly improve the performance of SLMs, thus eliminating the accuracy limitations of SLMs without involving the need of complex models like LLMs. In our approach, we integrate open-source models, LLaMa 3.1 (8B) and LLaMa 3.2 (3B) [3] with a retrieval system built on ChromaDB and benchmark them using MMLU Pro, a challenging multi-domain evaluation set. We construct a domain-specific knowledge base combining synthetic content (via Gemini API with grounding) and real legal text scraped from public repositories.

The goal is to determine whether this lightweight setup can achieve near-parity with larger models while operating locally. Our findings suggest that well-curated RAG pipelines can boost SLM performance, making them viable for real world, privacy focused applications. This work contributes to the broader push for decentralized, secure, and efficient AI systems that don't rely on centralized cloud infrastructure.

The main contributions of this paper are threefold. First, it presents a scalable pipeline for enhancing Small Language Models (SLMs) using Retrieval-Augmented Generation (RAG) with domain-specific datasets. Second, it evaluates this integration using the MMLU Pro benchmark across five complex domains, analyzing both performance gains and losses. Finally, it provides a domain-wise interpretation of where RAG helps, where it fails, and what improvements are needed to make retrieval more effective for grounded reasoning in local LLM deployments.

The remainder of this paper is structured as follows. Section 2 discusses related work on Large Language Models, RAG techniques, and benchmarking approaches. Section 3 describes our data pipeline, retrieval integration, prompt design, and experimental setup. Section 4 reports and analyzes the experimental results, focusing on domain-specific performance and the practical impact of RAG on LLaMa 3B and 8B models. Finally, Section 5 summarizes the key findings and highlights opportunities for future research.

II. Related Works

In today's era, demand for Large Language Models (LLMs) has rapidly increased. The models such as GPT-4, Claude and Gemini are pushing the boundaries of generative AI. These models are capable of performing an extensive variety of tasks like question answering, text summarization, and multilingual processing but are often hindered by the limitations of cost, energy consumption and privacy [4, 5]. In order to overcome these limitations of LLMs, Small Language Models are introduced. These models have less than 10 billion parameters and they can run on local hardware. Some examples may include Meta's LLaMa 3 models [3] and Google's Gemma 3 family [6]. Although SLMs excel with LLMs in cost, energy consumption and privacy but due to their limited training parameters, they produce less accurate results in tasks.

To eliminate the limitations of SLMs, we have explored Retrieval-Augmented Generation (RAG) technique. RAG combines a large language model with an external database that fetches relevant documents or facts from a knowledge base before generating output [4]. This framework helps RAG to predict more accurate results in complex tasks. Studies such as Zhang et al. [2] reveal that models using RAG perform significantly better in the fields of law and medicine. Trade-offs between different RAG techniques are analyzed in another study by Sakar and Emekci [4]. The techniques involved in this study are contextual re-ranking, generative re-ranking and multi-hop retrieval. The study reveals that combining different retrieval techniques often retrieves best results.

Going beyond traditional RAG, GraphRAG introduces knowledge graphs to the mix [7]. They involve multi-hop reasoning to traverse along the paths, thus meaningfully revealing more accurate and trustworthy information. Such models help in tasks like summarization based on queries and their relationships, indicating more accurate results.

We also examined another study involving MMLU-Pro benchmark [5] which extends the original MMLU [8] by adding more reasoning-focused questions. This makes it more suitable for testing improvements in reasoning and factual accuracy, particularly in smaller models.

Meanwhile, The concept of knowledge distillation was introduced by Hinton et al. [9]. It suggests that smaller models can achieve performance close to their larger counterparts by learning from the outputs of larger "teacher" models. This idea underpins several post-training methods, including Google's approach in Gemma 3 [6] which combines reinforcement learning and distillation to enhance reasoning.

A study analyzing four million Claude conversations by Anthorpic found that Generative AI usage today is concentrated in code related tasks [1]. This underlines the need for better grounding and domain adaptation for more sensitive or high-risk applications, such as healthcare or law.

In summary, while extensive work has gone into improving LLM accuracy and efficiency, there is still a gap in enabling small, locally-hosted models to perform competitively. This project builds on existing RAG and SLM research to demonstrate that with domain-specific datasets and retrieval optimizations, even SLMs can deliver reliable, privacy-friendly AI performance.

III. METHODOLOGY

A. Data Collection Pipeline

To enhance small language models (SLMs) using Retrieval-Augmented Generation (RAG), we developed a tailored data acquisition pipeline as shown in Fig. 1. Our goal was to supply the language model with deterministic, high-quality knowledge tailored for the MMLU Pro benchmark. The framework is composed of two modules: (1) synthetic content based on the questions, and (2) legal domain knowledge scraped from public archives.

A critical component of our RAG-based system was the construction of a high-quality knowledge base to support inference. To meet the complexity of the MMLU Pro benchmark, we combined synthetically generated documents along with scrapped content, ensuring coverage across science, history, and law while maintaining accuracy.

For synthetic data generation, we used the Gemini 2.0 Flash API with grounded search enabled. Over 500 topic subtopic pairs were generated to reflect the structure and depth of MMLU Pro domains. Examples include *Virology: Retroviruses and Genome Integration*, *Soviet Policies: Rural Education*, and *Organic Chemistry: Aromaticity and Reaction Stability*. Using a carefully designed prompt template, Gemini was instructed to return dense, standalone text blocks (around 200 to 500 words in length) containing definitions, examples, categories, and formulas. This approach ensured content suitable for precise embedding and fast lookup at inference time.

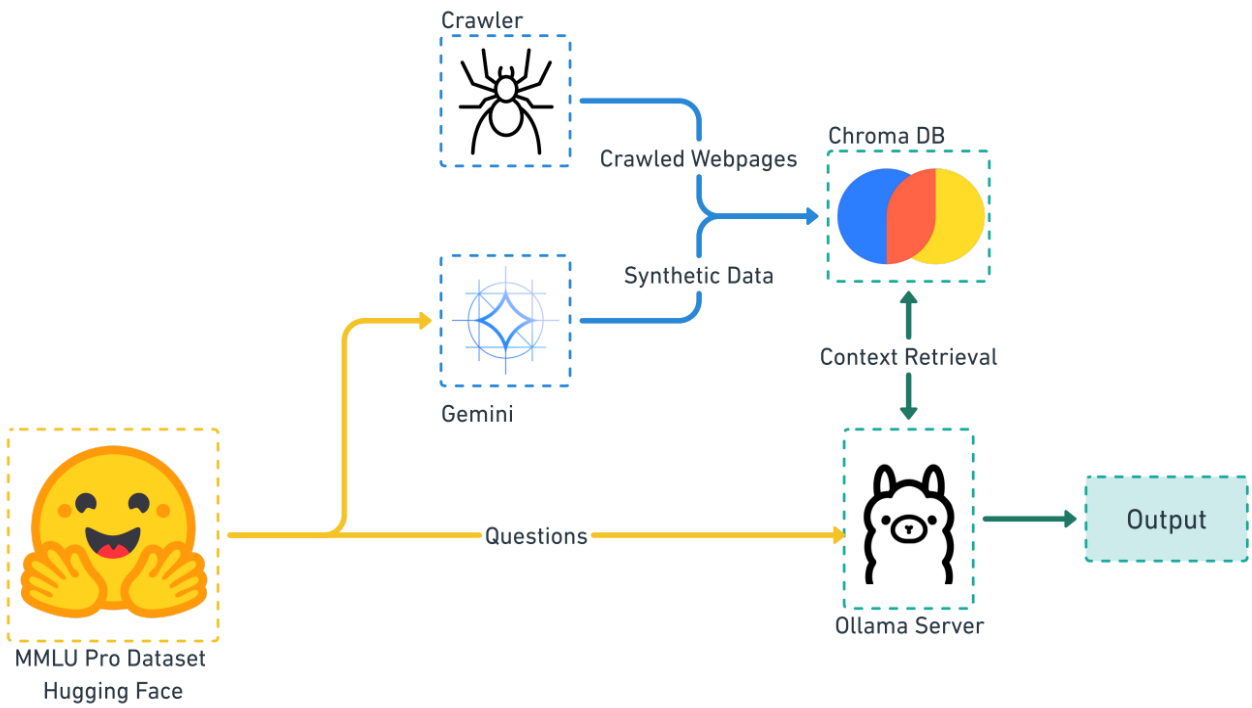

Fig.1. Benchmark Pipeline Design

For the legal domain, where factual accuracy and source credibility are absolutely critical, we opted for real data rather than synthetic generation as discussed before. We used the Crawl 4 AI Python library to scrape and process public content from the Cornell Legal Information Institute (LII). The pipeline performed multi-threaded crawling, HTML-to-markdown conversion, and content filtering using custom regex rules to strip boilerplate and retain only meaningful legal definitions and case references.

B. MMLU Pro Benchmark Overview

The Massive Multitask Language Understanding Pro (MMLU Pro) benchmark [5] is a well-established evaluation suite designed to measure both reasoning ability and factual recall across 14 broad categories and more than 50 specialized sub-domains. Compared to the original MMLU [8], it addresses several limitations by increasing the overall question difficulty through more multi-hop reasoning, expanding the answer choices from four to ten to lower the chances of guesswork, improving prompt stability with reduced variance across formats (about 2% compared to 5% previously), and widening its coverage to include fields such as biology, law, history, chemistry, and mathematics. For our experiments, we directly used the official mmlu_pro dataset available on HuggingFace, ensuring that all questions were properly loaded, filtered, and parsed for consistent evaluation across models.

C. Initial Hypothesis and Planning

The primary hypothesis driving this research was that integrating RAG should improve the accuracy of existing small language models. To test this claim, we designed a straightforward experimental setup. We evaluated two models, LLaMa 3.1 8B and LLaMa 3.2 3B using Ollama as our runtime environment. Each model was benchmarked in two phases: first without any retrieval support, and then with RAG-enabled context, while keeping the prompt format and question sets identical across both runs to ensure fair comparison. For performance measurement, we focused solely on accuracy. The models were required to output only the correct option, which we then directly compared against the official MMLU Pro answer key, avoiding any subjective interpretations.

D. Format for RAG and Non-RAG Prompts

To keep experiments controlled, both RAG and non-RAG models received nearly identical prompt structures, with the only difference being the inclusion of context. Here's how the prompt formats were constructed:

Without RAG (Baseline Prompt)

[Solve this question. Reply with just the index of the correct answer.

Ex: if option 0 is correct, reply: 0]

Question: {mmlu_question}

This prompt instructed the model to avoid longer responses and reply only with the answer index (0–9). This minimized parsing errors and eliminated noise, especially important in one-shot inference benchmarks.

With RAG (Augmented Prompt)

[Use the following context to solve this question. Reply with just the index of the correct answer.

Ex: if option 0 is correct, reply: 0]

Context: {retrieved_docs}

Question: {mmlu_question}

In this prompt, {retrieved_docs} represents 2-3 high-relevance documents retrieved from ChromaDB using cosine similarity with the question as the query. The embedding model used was all-MiniLM-L6-v2, chosen for its balance of speed and semantic precision.

This design made sure that the model grounded its answer in retrieved, domain-relevant information rather than relying so on its internal knowledge.

IV. RESULTS

Table 1: MMLU Pro Accuracy LLaMa 3B vs LLaMa 3B RAG

Domain	**LLaMa 3B**	**LLaMa 3B + RAG**
Law	5.00%	7.00%
Organic Chemistry	5.00%	23.53%
Foreign Policy	5.00%	10.00%
Medical Genetics	5.56%	9.26%
Geography	10.77%	10.77%
College Medicine	8.33%	14.58%
College Biology	12.66%	15.19%
High School Biology	16.43%	15.00%
History	13.25%	14.46%
World History	16.67%	16.67%

Table 2: MMLU Pro Accuracy LLaMa 8B vs LLaMa 8B RAG

Domain	**LLaMa 8B**	**LLaMa 8B + RAG**
Law	18.70%	19.20%
Organic Chemistry	23.50%	11.76%
US Foreign Policy	35.00%	45.00%
Medical Genetics	50.00%	35.19%
Geography	44.62%	41.54%
College Medicine	58.33%	33.33%
College Biology	43.04%	30.38%
High School Biology	48.57%	32.14%
US History	24.10%	25.30%
World History	41.67%	21.42%

A. Domain wise analysis

From the results in Tables 1 and 2, we can observe that the effectiveness of RAG depends on both domain and model size. The LLaMa 3B model consistently shows accuracy improvements when retrieval support is enabled, indicating that smaller models do benefit more from the added external knowledge. In contrast, the LLaMa 8B model displays mixed behavior, with some domains showing minor gains while others experience noticeable performance drops. This suggests that larger models are more sensitive to the quality and relevance of context, and poorly aligned information can actually interfere with the model's reasoning rather than help it.

In the Law domain, the performance gains were minimal. LLaMa 3B improved slightly from 5% to 7%, while LLaMa 8B saw an increase of only about 0.5%. This indicates that our retrieved content may not have contained the level of specificity or formal structure needed for legal reasoning, which typically depends on exact terminology and well-defined logical frameworks. As a result, RAG does not appear particularly effective here unless paired with more structured or symbolic retrieval approaches tailored for legal knowledge.

History shows mixed results. LLaMa 3B showed a small improvement in U.S. History but no gains in World History. LLaMa 8B saw a slight increase in U.S. History (from 24.10% to 25.30%) but a considerable drop in World History (from 41.67% to 21.42%). This decline likely stems from semantic mismatch or overly generic documents that introduced confusion. This supports prior findings by Sakar et al. [4] that poorly filtered context can interfere with a model's internal reasoning.

In Biology and Medicine, LLaMa 3B again benefited from RAG. College Medicine rose from 8.33% to 14.58%, and College Biology from 12.66% to 15.19%. These results suggest that smaller models can leverage external grounding effectively. LLaMa 8B however, dropped across all subdomains. College Medicine fell by 25%. These sharp declines imply that context injection can disrupt stronger models when the retrievals are noisy, redundant, or contradict the model's prior knowledge.

Organic Chemistry showed the most polarizing outcomes. LLaMa 3B improved by 18.5%, indicating strong compatibility between Gemini-generated content and the domain's deterministic structure. In contrast, LLaMa 8B dropped by nearly 12%, suggesting that the added context clashed with its internal knowledge possibly due to conflicting information or imprecise retrieval of data.

Geography showed little change overall. LLaMa 3B's performance remained essentially the same, while LLaMa 8B experienced a slight decline. This suggests that the models may already possess sufficient baseline knowledge for this domain, or alternatively, that the retrieved context did not provide enough detail or depth to meaningfully support geographic reasoning.

Overall, RAG significantly boosts the performance of smaller models like LLaMa 3B in knowledge-heavy tasks. However, for larger models like LLaMa 8B, context injection can be harmful without careful filtering and alignment of data. These findings highlight the importance of proper RAG strategies, where retrieval pipelines are tuned not only for relevance, but also for complementing the model's existing capabilities.

B. Findings and Analysis

The findings demonstrate that RAG is highly context sensitive. For smaller models like LLaMa 3B, RAG was consistently beneficial in four out of five domains. The most significant gains were in Organic Chemistry (+18.5%) and College Medicine (+6.25%), supporting the hypothesis that retrieved context can help SLMs [5].

In contrast, LLaMa 8B performed poorly, especially in biology and history. The performance drop in World History (–20.25%) and Organic Chemistry (–11.74%) shows that poorly aligned or unfiltered retrievals can degrade accuracy in stronger models.This could also indicate that once we scale from 3B to 8B with 5B additional parameters, we are essentially scaling to the point where any additional documents may not supplement info in a way which is unknown to the SLM.

RAG showed negligible impact in the fields of legal and geographic reasoning. This reveals that the parameters like quality, training in depth, context framework are also essential along with retrieving qualities.

C. Efficiency Analysis

While an improvement in accuracy is good, to be able to deploy such a system in more real world use case, we will need to demonstrate that the improved system in question:

1. Doesn't take too long to give out a meaningful answer
2. Does not add a significant memory overhead as to render it non feasible on resource constrained systems

To ensure this does not happen when such system is deployed in production use case we need to look at 2 factors:

1. Inference Time added as result of added context
2. Memory Overhead added by the vector database

First addressing the Memory Overhead, our vector database (ChromaDb) along with the particular dataset we used occupied 200Mb. This includes the Database and all the client libraries required to run this. While this may seem like a lot compared to the LLM themselves (1B parameter takes about 800Mb to store) So for our models and most other LLMs it will barely add around 10% for most use cases.

Now to address the inference time added by these systems, We ran performance benchmarks averaging the time it takes for our models to answer a question with or without context. They are available in Table 3

Table 3: Inference Time of LLMs

Modal	Context Added	Inference Time
LLaMa 3B	No	7.3 sec
LLaMa 3B	Yes	7.7 sec
LLaMa 8B	No	8.1 sec
LLaMa 8B	Yes	8.4 sec

From Table 3 we can tell that RAG setup adds around 400Ms to our response which once again means that our system is now around 5% slower which is not that significant considering the entire processing is now happening on a single personal laptop as opposed to the cloud setups.

Another thing to consider here is the effects of alternatives as well, adding 1B more parameters tend to increase the inference time by around 300Ms and adds around 800Mb or more to the overall overhead, In these cases our setup tends to be more efficient as compared to just using a larger model

From these observations we can conclude that creating such a system is feasible in production environments as the added overheads are better than the alternatives and are also giving out better, more grounded responses.

V. CONCLUSION

This study investigated how Retrieval Augmented Generation (RAG) can improve the factual accuracy of small language models, specifically LLaMa 3B and 8B, using the MMLU Pro benchmark. We evaluated whether retrieving domain-specific context at inference time helps mitigate hallucinations and enhance reasoning performance. Results showed that RAG consistently improved performance for LLaMa 3B across most domains, most notably in Organic Chemistry, College Medicine, and Law highlighting that smaller models benefit significantly from external grounding.

In contrast, the LLaMa 8B model produced mixed outcomes, with certain domains like Organic Chemistry and World History showing reduced accuracy after applying RAG. This suggests that larger models, which already have stronger internal knowledge, might be affected by irrelevant or misaligned data. Law and Geography showed negligible changes across both models. These findings emphasize that while RAG is promising for smaller models in privacy conscious deployments, its success depends heavily on retrieval quality, domain , and prompt design especially for scaling up to more capable architectures

To make the LLaMa 3B model competitive with larger models like the 8B variant and beyond, future work should focus on building more directed and deeply grounded domain-specific datasets. Enhancing the quality of retrieval data through better document curation, query reformulation, and context-aware re-ranking can help improve data quality. Additionally, incorporating multi-hop retrieval, graph-based knowledge structures, or self-distillation from larger teacher models can significantly enhance the reasoning capabilities of small models while maintaining their deployability on local, low-resource hardware

REFERENCES

[1] Handa, K., Tamkin, A., McCain, M., Huang, S., Durmus, E., Heck, S., ... & Ganguli, D. Which Economic Tasks are Performed with AI? Evidence from Millions of Claude Conversations.
[2] Li, J., Yuan, Y., & Zhang, Z. (2024). Enhancing llm factual accuracy with rag to counter hallucinations: A case study on domain-specific queries in private knowledge-bases. arXiv preprint arXiv:2403.10446.
[3] Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., ... & Ganapathy, R. (2024). The llama 3 herd of models. arXiv preprint arXiv:2407.21783.
[4] Şakar, T., & Emekci, H. (2025). Maximizing RAG efficiency: A comparative analysis of RAG methods. Natural Language Processing, 31(1), 1-25.
[5] Wang, Y., Ma, X., Zhang, G., Ni, Y., Chandra, A., Guo, S., ... & Chen, W. (2024, June). Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. In The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track.
[6] Team, G., Kamath, A., Ferret, J., Pathak, S., Vieillard, N., Merhej, R., ... & Iqbal, S. (2025). Gemma 3 technical report. arXiv preprint arXiv:2503.19786. 52
[7] Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A., ... & Larson, J. (2024). From local to global: A graph rag approach to query-focused summarization. arXiv preprint arXiv:2404.16130.
[8] Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., & Steinhardt, J. (2020). Measuring massive multitask language understanding. arXiv preprint arXiv:2009.03300.
[9] Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the knowledge in a neural network. arXiv preprint arXiv:1503.02531.